

Static Resume Website using Amazon S3 & CloudFront (AWS)

Project Summary

This project focuses on designing and deploying a **static resume website** using **Amazon Web Services (AWS)**. The website was hosted on **Amazon S3** using static website hosting and distributed globally through **Amazon CloudFront** to improve performance, reliability, and security.

The project provided hands-on experience with **cloud storage, CDN configuration, access control, error handling, pricing plans, and cost management**. Special attention was given to understanding **AWS service behavior**, troubleshooting real-world errors, and ensuring **safe resource cleanup** to avoid unexpected billing.

Project Objectives

- Host a static website using Amazon S3
- Configure CloudFront for global content delivery
- Understand static website hosting concepts
- Implement secure access between CloudFront and S3
- Handle common CloudFront and S3 errors
- Learn AWS pricing plan behavior and safe cleanup

Architecture Overview

- **Amazon S3** stores static website files
- **Amazon CloudFront** acts as a CDN in front of S3
- Users access the website through the CloudFront domain
- CloudFront fetches content securely from S3

Core Concepts Explained

- ◇ Amazon S3 (Simple Storage Service)

Amazon S3 is an object storage service used to store static files such as:

- HTML

- CSS
- Images

Static Website Hosting in S3

- Allows S3 buckets to behave like a web server
 - Requires:
 - Index document (e.g., `index.html`)
 - Optional error document
 - Uses a **website endpoint**, not a REST API endpoint
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◇ Amazon CloudFront

CloudFront is a **Content Delivery Network (CDN)** that:

- Caches content at edge locations worldwide
- Reduces latency and improves performance
- Adds HTTPS and security protections
- Controls how requests are forwarded to the origin

CloudFront Distribution

A **distribution** defines:

- Origin (S3 bucket)
 - Cache behavior
 - Security rules
 - Default root object
 - Error handling
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◇ Default Root Object

- Specifies the file returned when a user accesses `/`
 - Example: `index.html`
 - Without this, CloudFront returns **AccessDenied** or **403 errors**
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◇ Error Handling

- CloudFront forwards errors from S3 if misconfigured
- Common errors:
 - **AccessDenied**: Bucket permissions or missing index file
 - **403 Forbidden**: Static hosting not enabled or private access issue
 - **404 Not Found**: File not present in S3

Challenges Faced & Solutions

AccessDenied Error from CloudFront

Issue:

- CloudFront returned an XML AccessDenied error

Cause:

- Default root object not configured
- Static website hosting not enabled initially

Solution:

- Enabled static website hosting in S3
- Set `index.html` as Default Root Object in CloudFront

Confusion Between S3 REST Endpoint and Website Endpoint

Issue:

- CloudFront was pointing to the wrong S3 endpoint

Cause:

- S3 REST endpoint behaves differently than website endpoint

Solution:

- Understood endpoint types and ensured correct origin configuration

CloudFront Deletion Restriction

Issue:

- CloudFront distribution could not be deleted immediately

Cause:

- Distribution was subscribed to a **Free pricing plan**
- AWS enforces a billing cycle before deletion

Solution:

- Disabled the distribution
- Cancelled the pricing plan
- Scheduled deletion after billing cycle end

Cost & Billing Anxiety**Issue:**

- Concern about unexpected AWS charges

Solution:

- Emptied and deleted the S3 bucket immediately
- Ensured CloudFront was disabled
- Verified Free plan usage (0%)
- Confirmed AWS does not charge for deleted resources

Broken Origin Scenario**Issue:**

- S3 bucket deleted while CloudFront still existed

Impact:

- CloudFront origin became invalid

Understanding:

- Disabled CloudFront does not fetch content
- AWS does not penalize or charge for broken origins

Security Considerations

- S3 bucket access restricted to CloudFront
- No public bucket exposure
- HTTPS enabled via CloudFront
- No custom domain or certificates used

Cleanup Strategy

- Disabled CloudFront distribution
- Emptied S3 bucket
- Deleted S3 bucket
- Scheduled CloudFront deletion post billing cycle

No AWS charges incurred

Key Learnings

- How static website hosting works in S3
- How CloudFront distributions function
- Real-world AWS error troubleshooting
- Pricing plan lifecycle and deletion constraints
- Importance of cost-aware cloud design

Conclusion

This project strengthened practical knowledge of AWS core services and exposed real-world cloud challenges involving configuration, security, and billing. It highlights the importance of understanding **service dependencies**, **error messages**, and **cleanup processes** when working in cloud environments.

